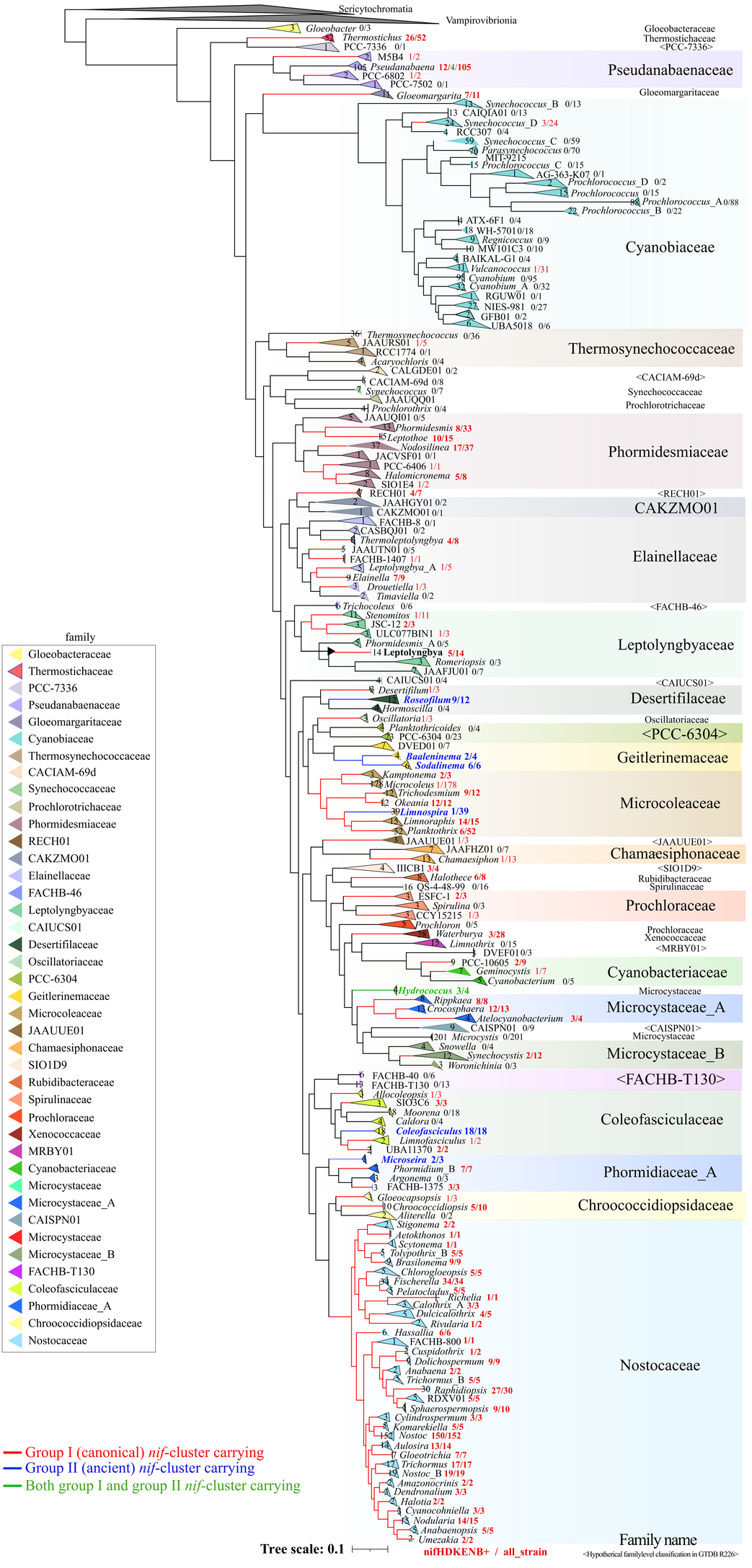

Supplementary Figure S9. Cyanobacterial phylogenetic tree at genus level showing the distribution of canonical (Group I) and ancestral (Group II) *nif* carrying lineages. A total of 2,716 representative strains from GTDB R226 were analyzed, excluding GTDB hypothetical genera. Phylogenetic inference was performed using the approximately ML method implemented in FastTree (Price et al., 2009), with the WAG substitution model and gamma-distributed rate heterogeneity among sites. The analysis was based on approximately 5,000 highly conserved, gap-free sites derived from the 120 core genes defined by GTDB (Parks et al., 2018). Family-level taxonomic classifications defined in R226 are highlighted with background colors. The red, blue and green digits indicate the number of strains carrying canonical (Group I), ancestral (Group II) and both *nif* genes, respectively; relative to the total number of strains in each genus identified by Nif-Finder. The full-size figure is available from the Nif-Finder repository.